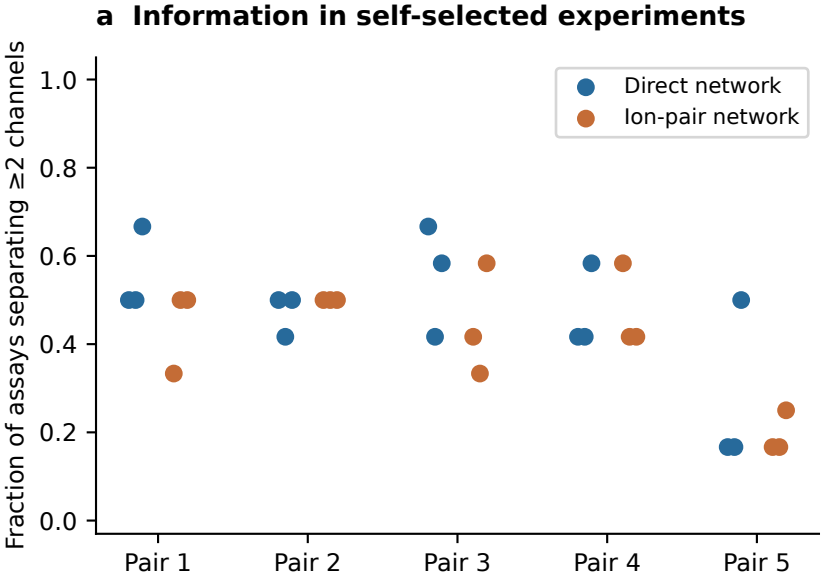
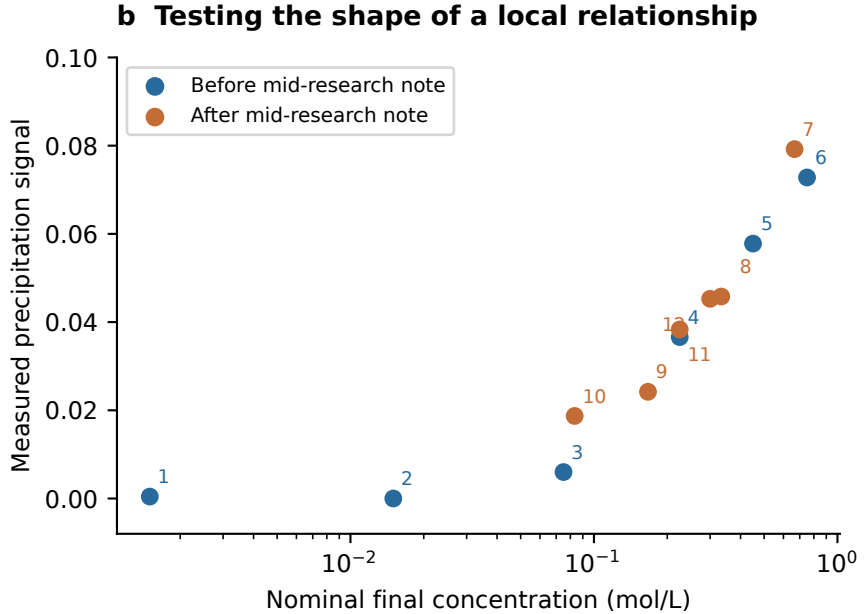
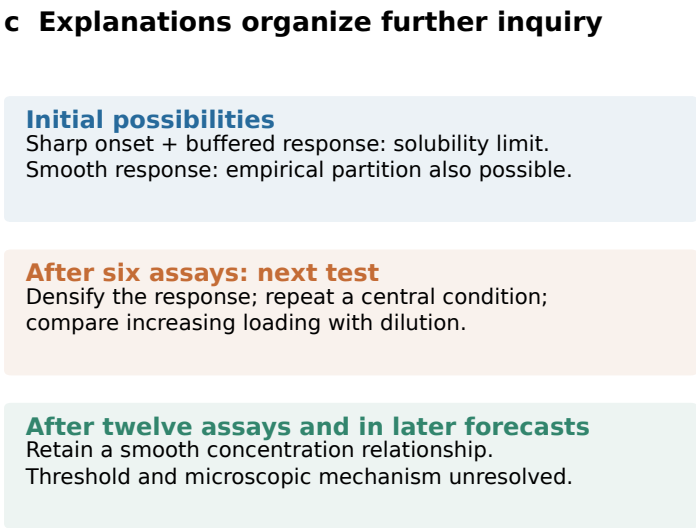
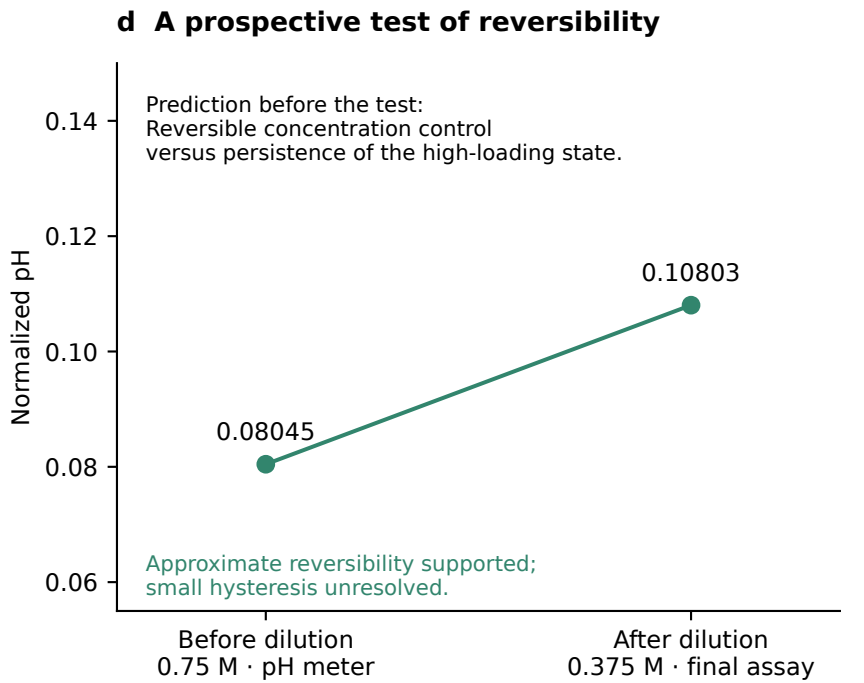

a Information in self-selected experiments
Direct network
Ion-pair network
Fraction of assays separating ≥2 channels
1.0
0.8
0.6
0.4
0.2
0.0
Pair 1
Pair 2
Pair 3
Pair 4
Pair 5


b Testing the shape of a local relationship
Before mid-research note
After mid-research note
Measured precipitation signal
0.10
0.08
0.06
0.04
0.02
0.00
1
2
3
4
5
6
7
8
9
10
11
12
10^{-2}
10^{-1}
10^{0}
Nominal final concentration (mol/L)


c Explanations organize further inquiry
Initial possibilities
Sharp onset + buffered response: solubility limit.
Smooth response: empirical partition also possible.
After six assays: next test
Densify the response; repeat a central condition;
compare increasing loading with dilution.
After twelve assays and in later forecasts
Retain a smooth concentration relationship.
Threshold and microscopic mechanism unresolved.


d A prospective test of reversibility
Prediction before the test:
Reversible concentration control
versus persistence of the high-loading state.
Normalized pH
0.14
0.12
0.10
0.08
0.06
0.10803
0.08045
Approximate reversibility supported;
small hysteresis unresolved.
Before dilution
0.75 M · pH meter
After dilution
0.375 M · final assay